

TET~CRAFT: An Architecture of Unified Molecular Physics

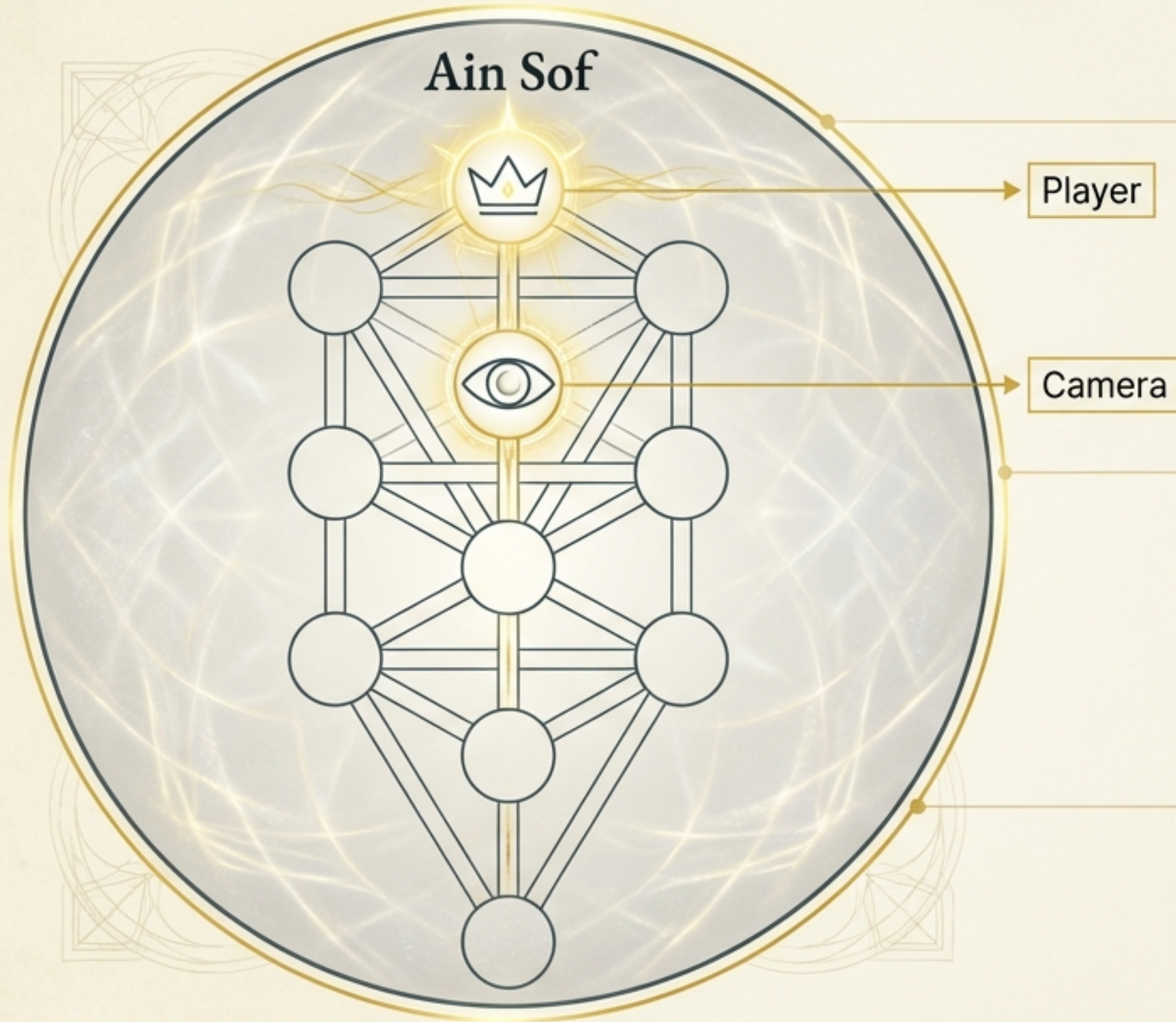


A molecular geometry simulation modeling a 'kleiniverse' through the interaction of tetrahedral units (TETs).

A unified physics environment designed to explore the intersection of matter, consciousness, and the genesis of complex systems.

This is a tour of its blueprint: a dynamic Tree of Life, where divine intent is manifested in physical law.

The Kleinverse: From Boundless Potential to Perceived Reality



Ain Sof (The Infinite)

The simulation exists within a shimmering sphere of past events—the **4D Past Projection**. This is Ain Sof, the undifferentiated potential from which all forms emerge and to which they return.

The Player as Keter (The Crown)

The Player is the divine will, the “prime observer” missing from quantum theory. They initiate creation from above the tree without direct intervention.

The Camera as Da'at (Knowledge)

The Camera is the invisible sephirah where the light of understanding enters the system, making the kleinverse knowable.

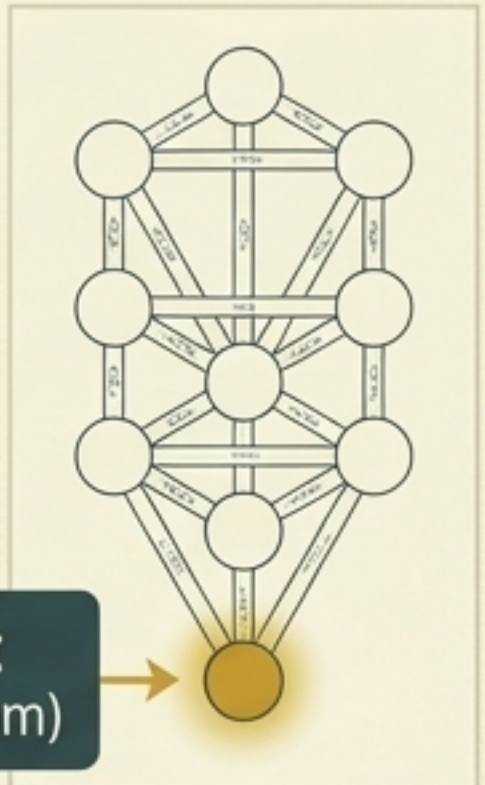
The TET: A Kingdom Crystallized from the Void

Malkhut (The Kingdom):
Each TET is a "Kingdom" in miniature.

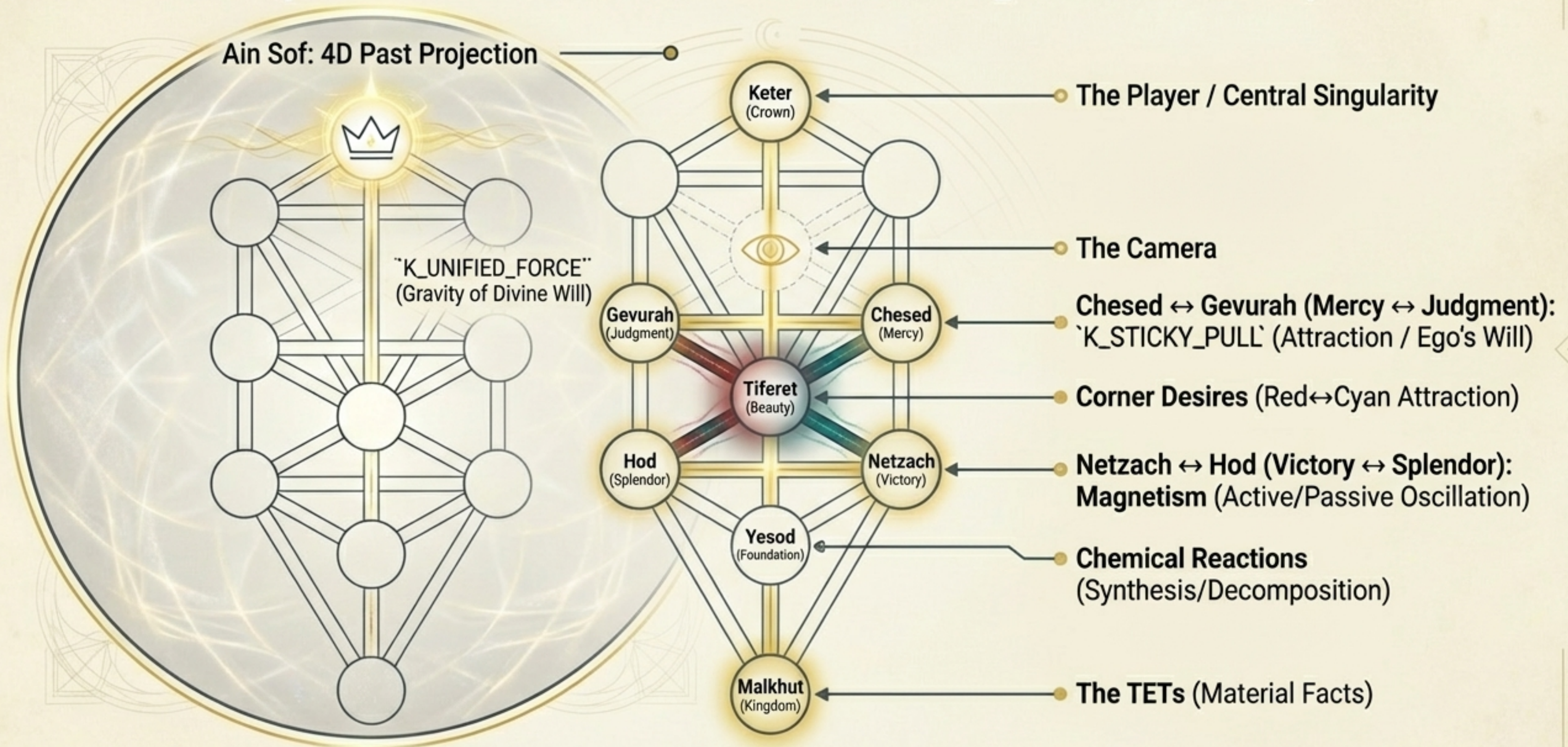
A Building Block of Reality:
The TET is the most basic 3D structure, a "fact" crystallized from the void. It is the destination for the descending divine light.

The Foundation of Understanding:
Connecting these individual "facts" (TETs) allows them to cohere into a larger (mis)understanding of reality.

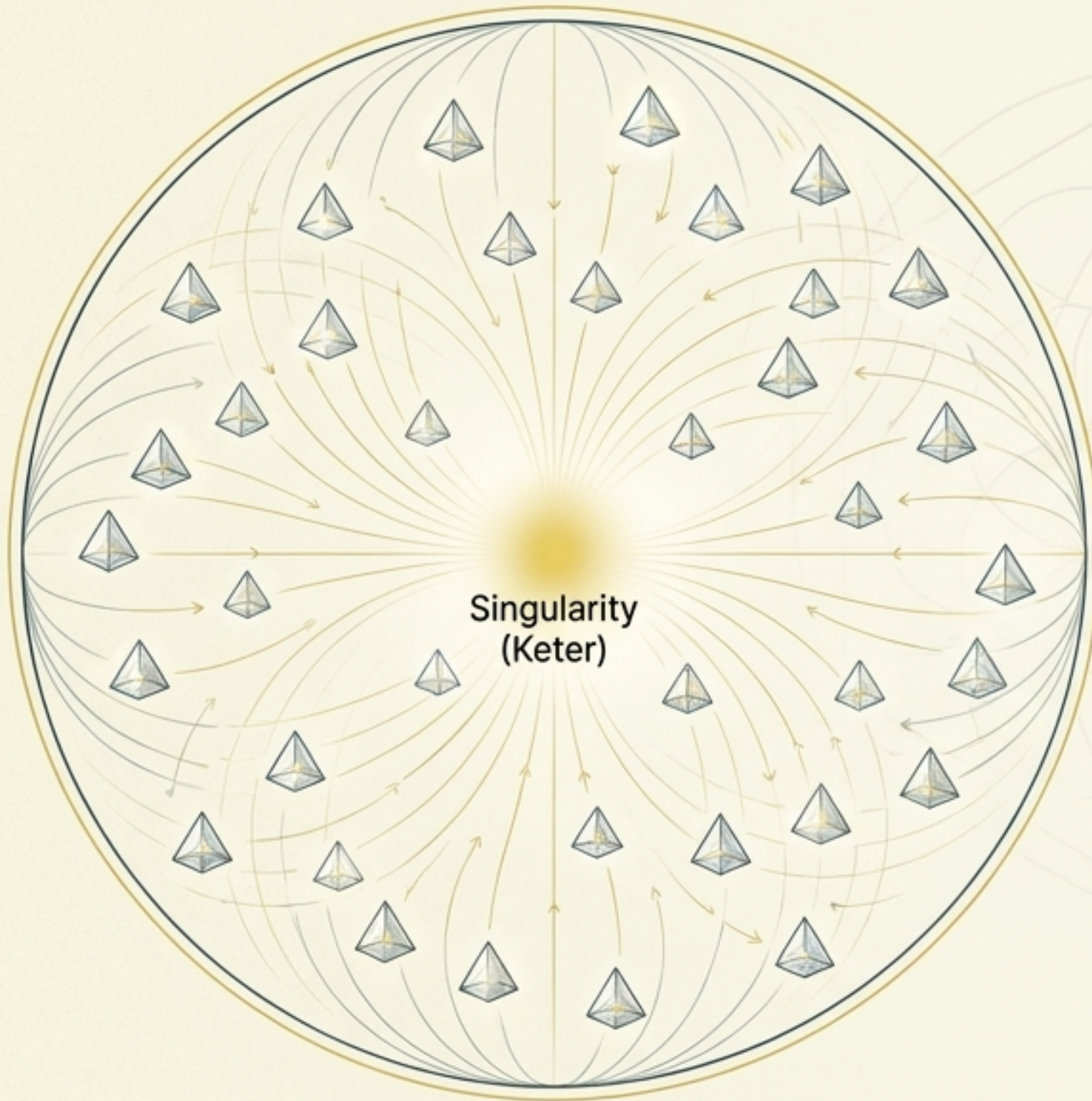
The Act of Creation:
Pressing the space bar creates a new TET—a new fact to be misunderstood and eventually cohered into the whole.



The Dynamic Tree of Life: A Blueprint for Physics



God's Will as Physical Law: The Unified Force



Metaphysical Principle

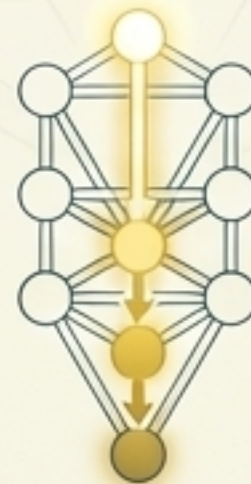
Keter to Malkhut. The descending light of God's will that structures existence and prevents reality from dissolving into chaos.

Physical Implementation

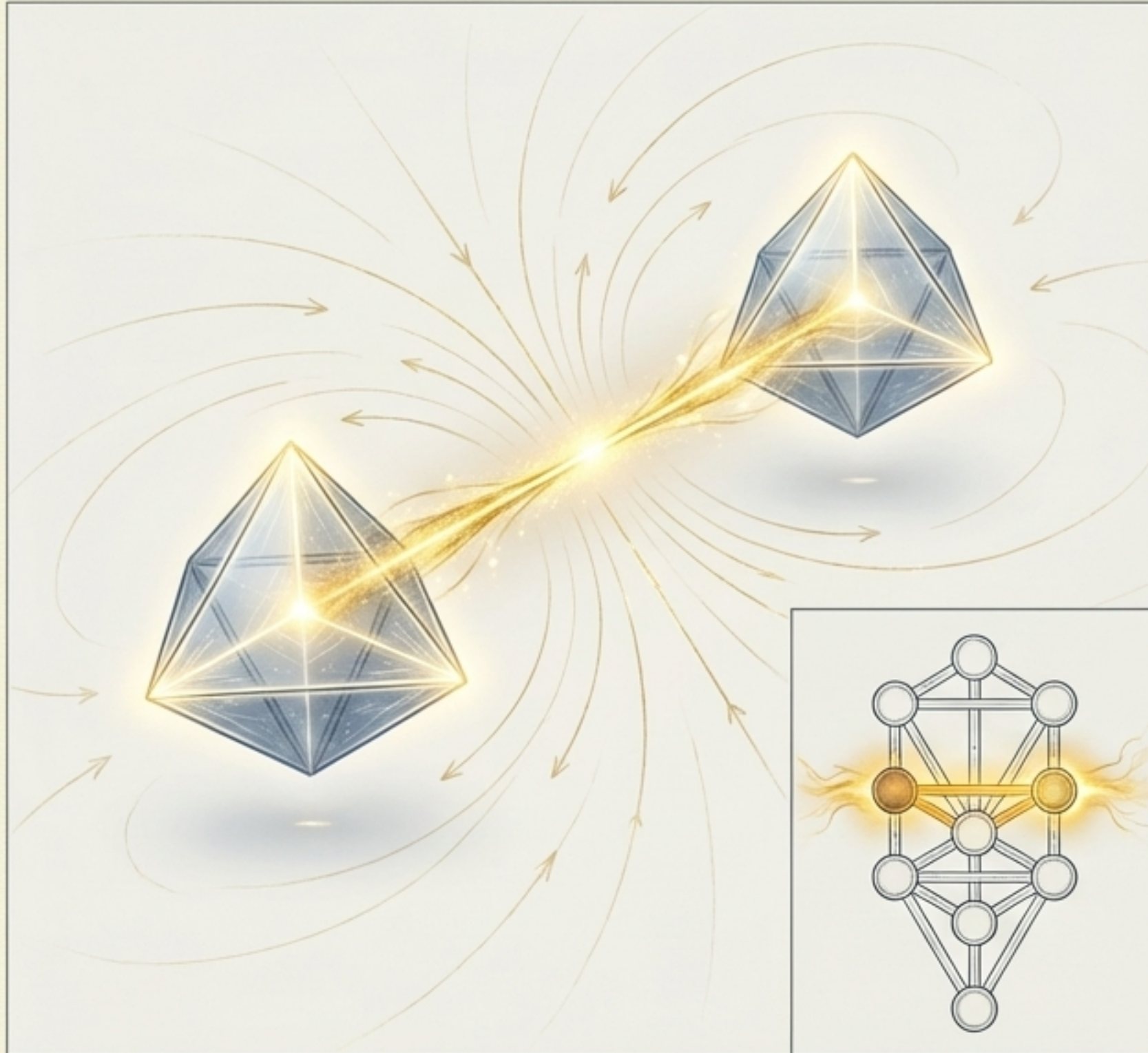
- ``K_UNIFIED_FORCE``.
- A persistent gravitational pull toward a central singularity.
- **Value:** ``0.00001`` ("Tiny in the extreme").
- Acts as a "free will override," gently nudging the system toward an optimal perspective.

Analogy

- A vast, silent whirlpool in a dark ocean. The ``K_UNIFIED_FORCE`` is the steady, almost imperceptible pull ensuring every drop eventually returns to the source.



The Ego's Attraction: A Dance of Mercy and Judgment



Metaphysical Principle

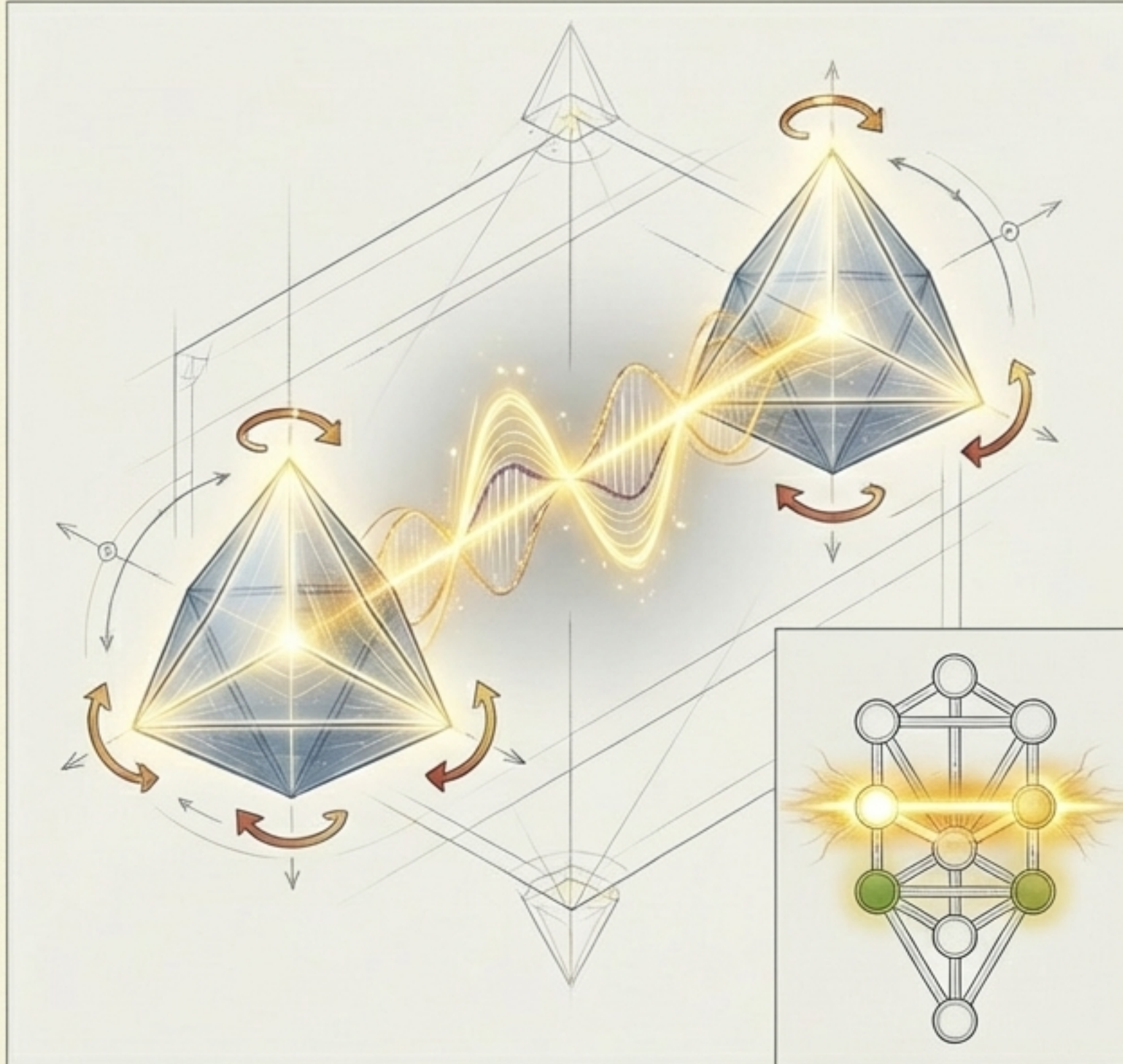
Chesed ↔ **Gevurah**. The creative tension between expansion (pulling together) and limitation (maintaining separation) that generates all relationships.

Physical Implementation

`K\_STICKY\_PULL` (or `K\_PULL`).

- An “image” of the unified force, but powered by the **ego's will**. It is “less perfect, less understanding.”
- **Value:** 50.0.
- Refined as a **Lennard-Jones composite potential**: an attraction constant (k_-/r^2) balanced by a repulsion that only activates after “sustained proximity over time,” where stabilization is faster with greater “coherence” (understanding).

The World's Rhythm: Magnetism as Oscillation



Metaphysical Principle

Netzach ↔ **Hod**. The eternal oscillation between active (**Victory**) and passive (**Splendor**) energies that sustains the world.

Physical Implementation

Magnetism.

- **Torque-Based:** Uses a spin alignment term to align TETs, mimicking dipole-dipole interactions, rather than a linear pull. Strength falls off at $1/r^3$.
- **Energy Exchange:** Bonded magnetic pairs exchange "emptiness" (1 - battery level"), creating a dynamic **Battery Oscillation**.

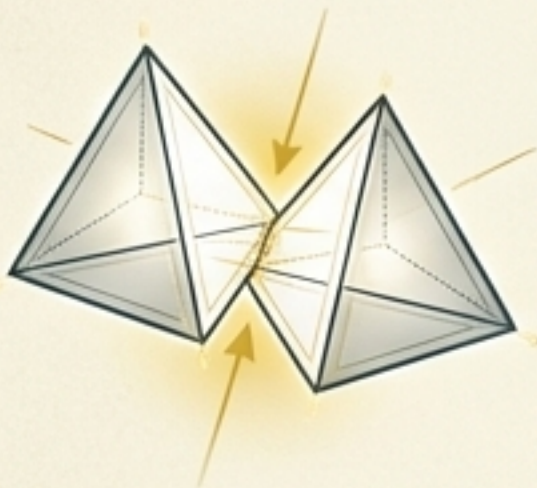
Analogy

- If the Unified Force is the ballroom floor and Sticky Pull is dancers holding hands, Magnetism is the rhythmic music forcing them to shift and move, ensuring the dance continues.

Geometry as Destiny: The Emergence of Chemistry

Multi-Face Locking

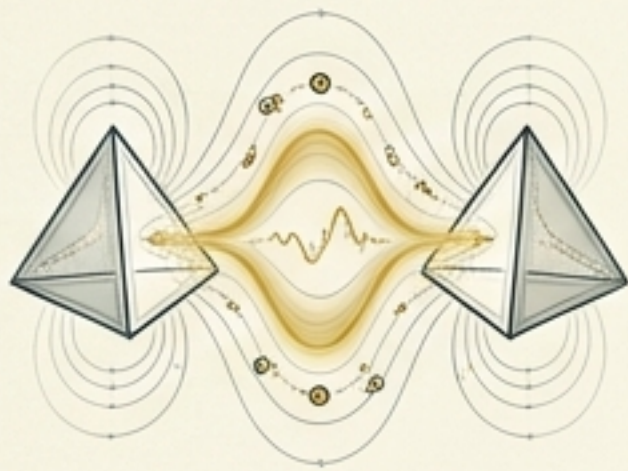
White ↔ White = Carbon



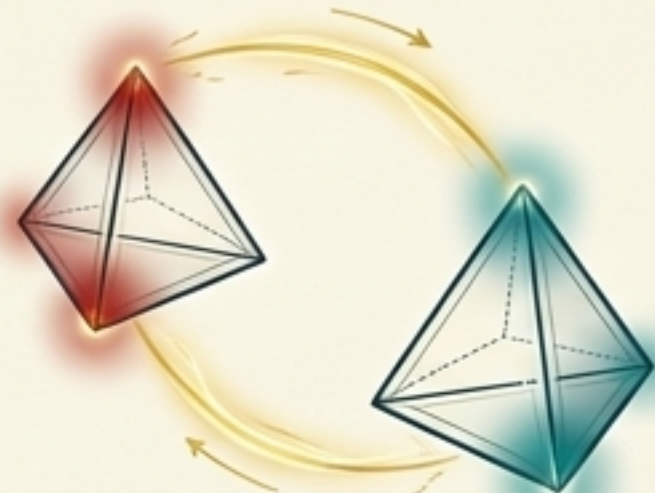
White ↔ White = Carbon

Each lock adds +0.25
magnetic strength.

Battery Oscillation

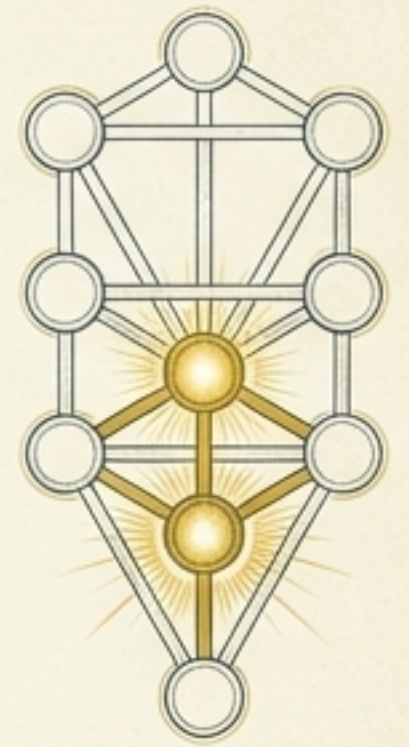
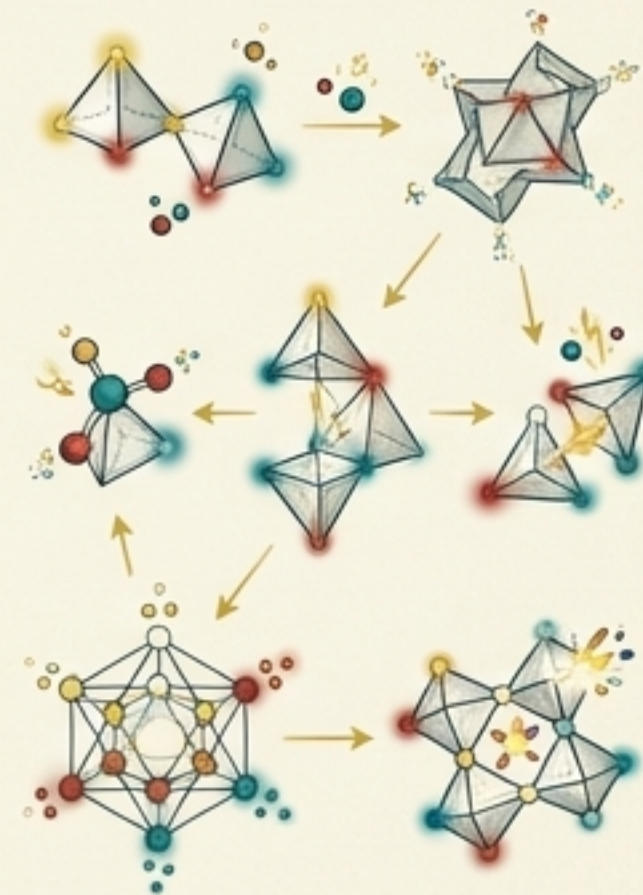


Corner Desires (Tiferet - Beauty)

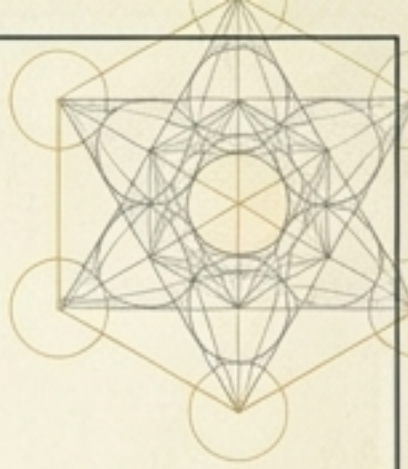


Red corners seek Cyan corners
("spooky action at a distance"). This
attraction between "contradictory
colours" creates a harmonious
union (Hydrogen).

Chemical Reactions (Yesod - Foundation)



The Price of Existence: Entropy Leakage



A Thermodynamic Constraint

Entropy Leakage is a continuous loss of battery power for every TET, a 'tax' on existence and complexity.

Mechanics

- **Decay:** Every TET loses battery over time and through interaction.
- **Starvation:** At 0% battery, bonds weaken.
- **Collapse:** If the population drops too low due to energy depletion, the simulation can collapse to an earlier era.

The Metabolic Imperative

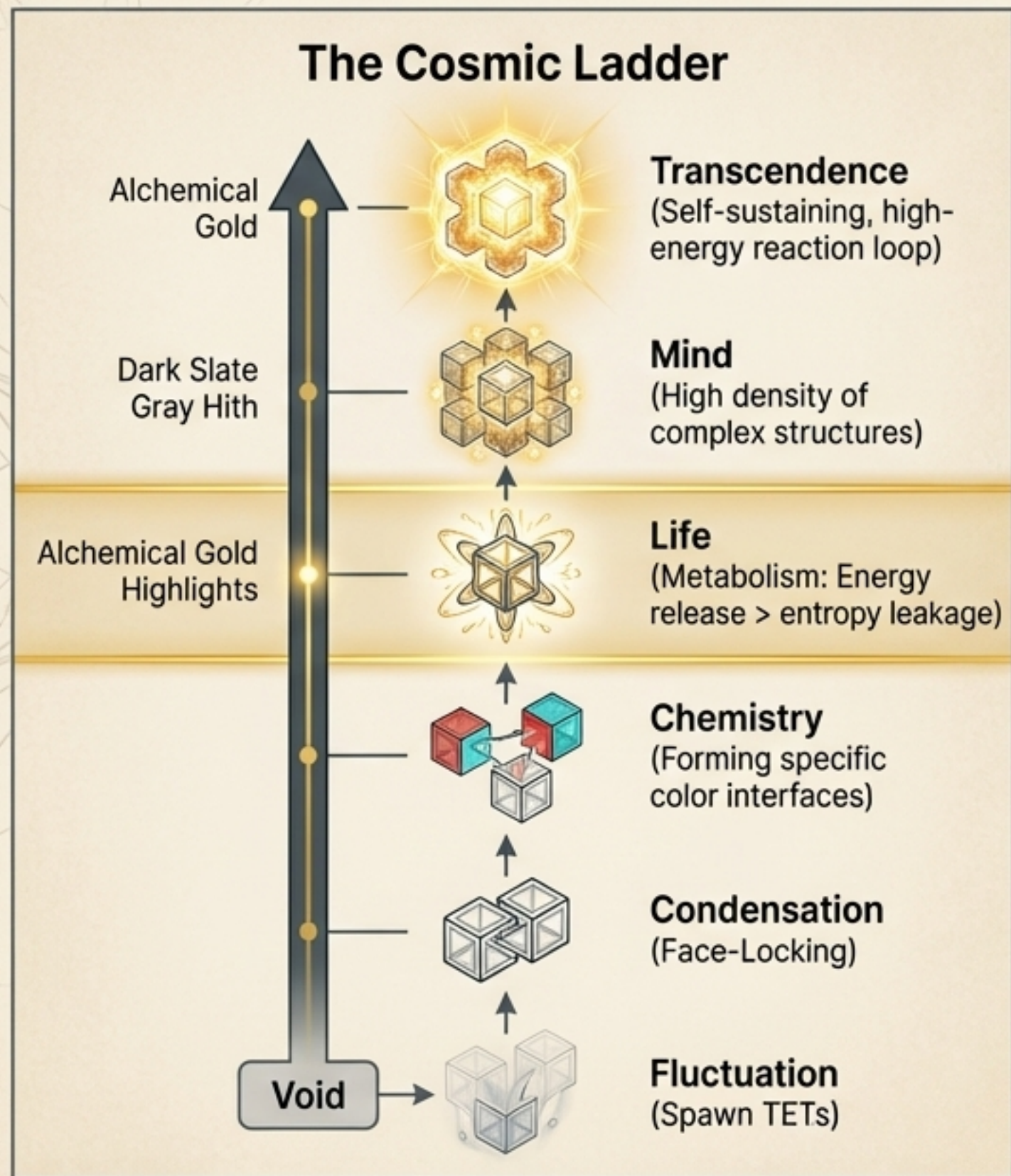
The "Great Filter" to reach the "Life" era is to achieve a state where **energy release from reactions is greater than entropy leakage**.

Analogy

The simulation is a leaky bucket. To keep it full enough to support complex structures, you must find ways to pour water in faster than it leaks out. This is **metabolism**.



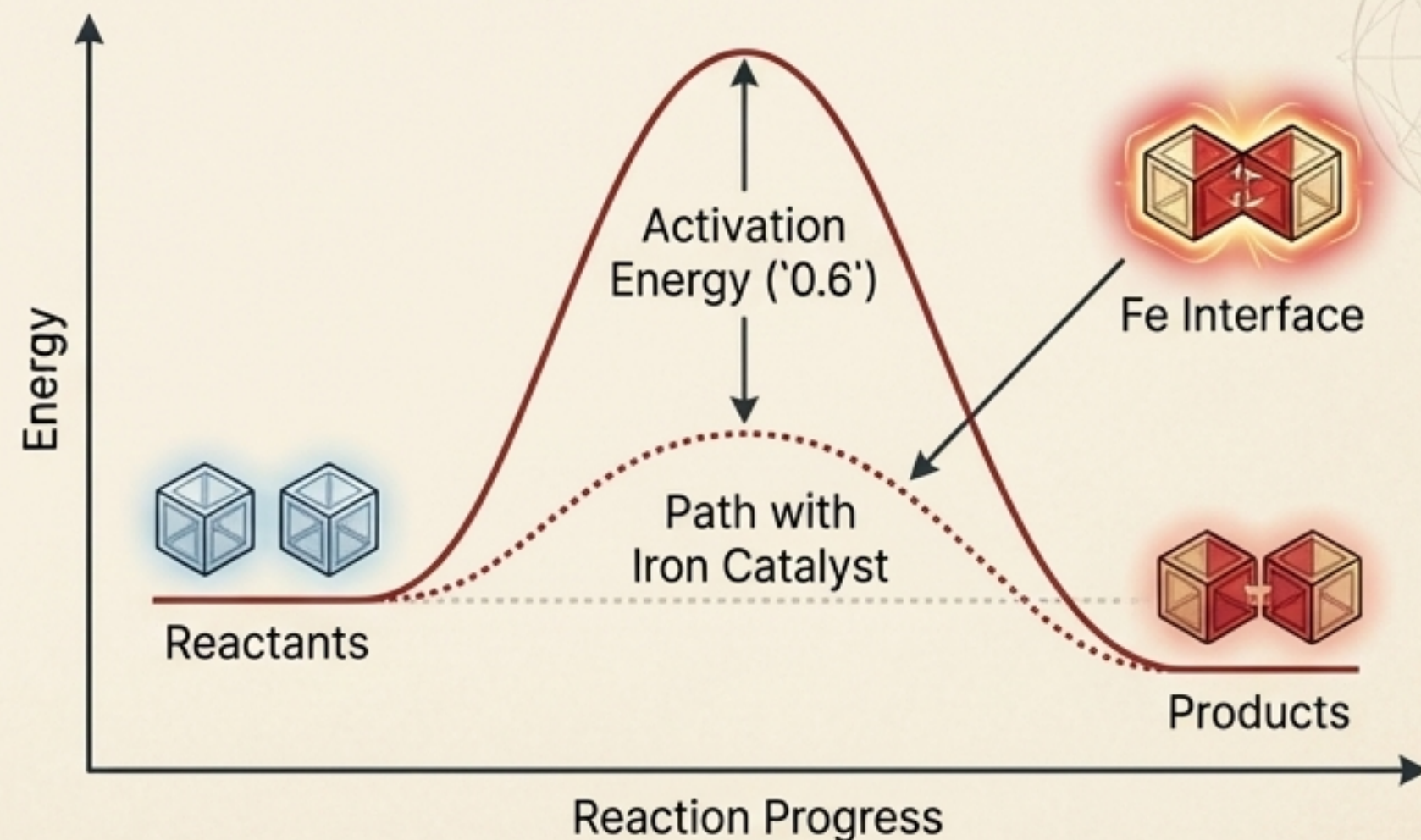
The Cosmic Ladder: Progress Through Complexity



The Great Filter: Activation Energy

Reactions require Activation Energy to occur; "cold" molecules are inert. Base energy is `0.6`.

The Catalyst Strategy: An Iron interface (Red ↔ Red faces) creates a magnetic dipole that **lowers the activation energy barrier**, making reactions possible at lower temperatures.



Time as Misunderstanding, Lies as Delay

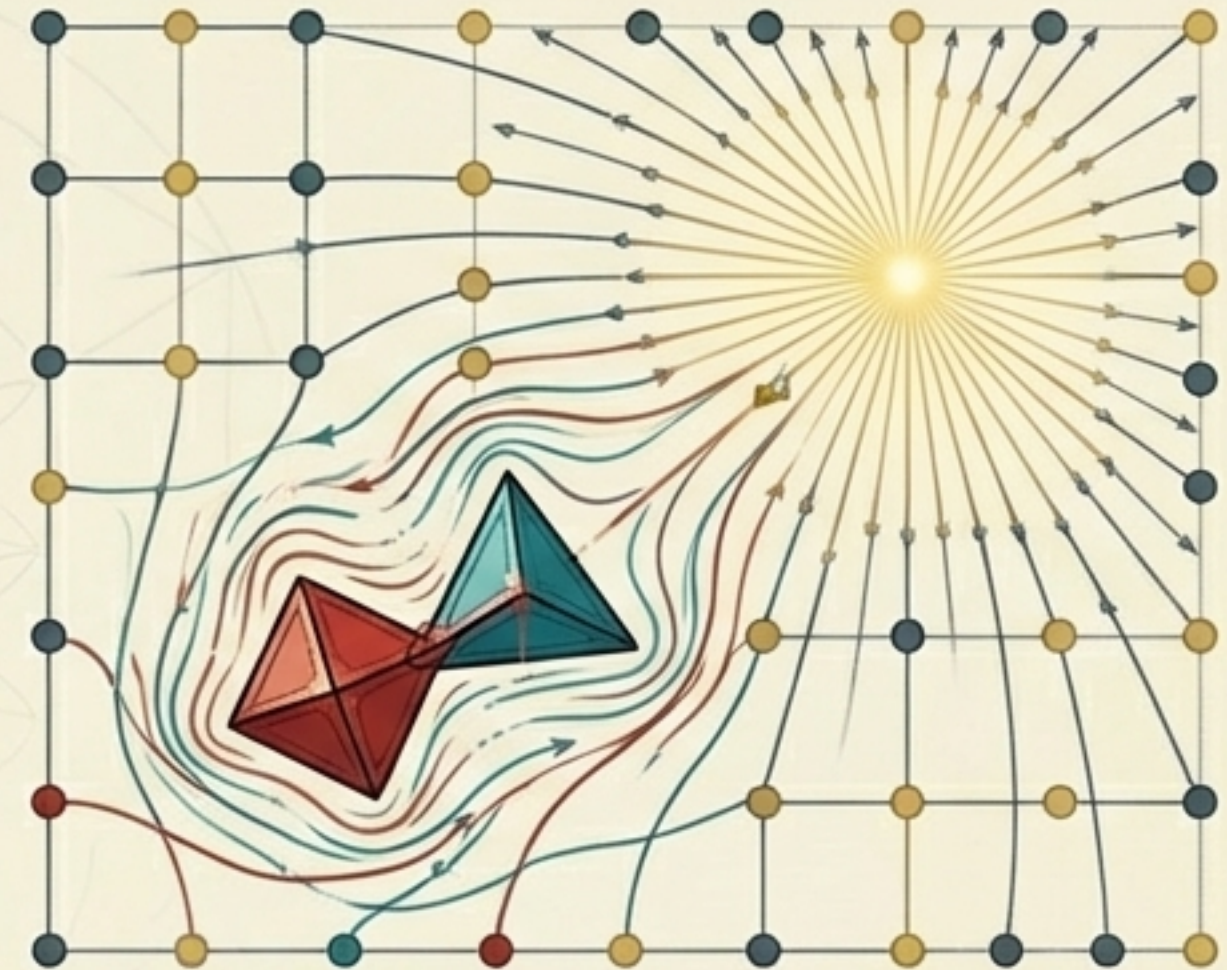
Time as a Product of Perspective



Time is essentially the result of misunderstanding. It emerges from different perspectives and memories, allowing for discovery instead of immediate recognition.

The duration of the effort—the interval between seeing a fragment and discovering where it fits—is what we perceive as Time.

Lies as the Resistance to Unity



Lies are the only way to delay God's will (the natural pull toward central collapse and cohesion).

A lie is a warping of the view and model of reality. It doesn't change the facts (TETs), but it creates a local warp that stops other pieces from clicking together.

A Cathedral of Code, A Universe in a Machine

TET~CRAFT presents a unified architecture where every physical law is a **manifestation** of a **metaphysical blueprint**.

Each **TET** is a **brick**, its interactions governed by forces drawn from the **Sefirot**, building a complete house of knowledge within the **infinite** potential of Ain Sof.



A Final Speculation

The simulation demonstrates the computational requirements for reality. It would take ~300 Earth-like planets to simulate the Milky Way. Or perhaps...

...the Milky Way itself is not solid matter, but a **5D past projection** opposing the nearest black hole.